NOT MEASUREMENT SENSITIVE

MIL-STD-361-6(TM)
15 JULY 1992

SUPERSEDING
(See 6.5.)

MILITARY STANDARD

MANUALS, TECHNICAL:

INTRODUCTORY INFORMATION WITH THEORY OF OPERATIONS REQUIREMENTS

AMSC A6755

AREA TMSS

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FOREWORD

1. This military standard is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense (DOD).

2. Beneficial comments (recommendations, additions, and deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, USAMC Materiel Readiness Support Activity, ATTN: AMXMD-EP, Lexington, Kentucky 40511-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

3. This document is one part of an 11-part bookform standard to be used in the preparation and acquisition of technical manuals that are essential to support (operate and maintain) the various types of equipment and weapon systems within the Department of the Army.

4. This document covers the introductory information with theory of operations requirements for the preparation of technical manuals.

5. Other parts of this multipart standard are:

MIL-STD-361-1	-	Writing Style and Format Requirements.
MIL-STD-361-2	-	Comprehensibility Requirements.
MIL-STD-361-3	-	Requirements for Graphics.
MIL-STD-361-4	-	Quality Assurance Requirements.
MIL-STD-361-5	-	Requirements for Packaging.
MIL-STD-361-7	-	Operator and Preventive Maintenance Instructions.
MIL-STD-361-8	-	Maintenance Instructions.
MIL-STD-361-9	-	Troubleshooting Procedures.
MIL-STD-361-10	-	Lubrication Instructions.
MIL-STD-361-11	-	Repair Parts and Special Tools List (RPSTL).

6. By incorporating standardized and service-unique technical manual requirements, this document and its dash-numbered companion parts, along with the general acquisition specification MIL-M-63008, Acquisition/Preparation Requirements, support the acquisition of technical manuals.

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1. SCOPE

1.1 Scope. This document establishes the content and procedural requirements for preparation of introductory information and theory of operations requirements for technical manuals (TMs), revisions, supplements, and changes.

1.2 Purpose. This document supplements the various content requirements in the other parts to this 11-part standard for specific types of technical manuals and related publications and does not alone deliver any technical data. (NOTE: The figures used in this document are examples only. The text of this document takes precedence over the figures.)

1.3 Applicability. The applicability of the requirements contained in this document is governed by the equipment, systems, or weapon systems for which the technical manuals are being developed. This document is applicable for use by the Department of the Army.

1.4 Selective application and tailoring. This document contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in this document are the responsibility of the contracting activity and shall be accomplished through the use of MIL-M-63008. The applicability of some requirements is also designated by one of the following statements:

- unless requirement is specifically excluded by the contracting activity; or
- as/when specified by the contracting activity.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2.)

SPECIFICATIONS

MILITARY

MIL-M-63008	-	Manuals, Technical: Acquisition/ Preparation Requirements.

STANDARDS

MILITARY

MIL-STD-12	-	Abbreviations for Use on Drawings, and in Specifications, Standards and Technical Documents.
MIL-STD-361-1	-	Manuals, Technical: Writing Style and Format Requirements.
MIL-STD-361-4	-	Manuals, Technical: Quality Assurance Requirements.
MIL-STD-361-5	-	Manuals, Technical: Requirements for Packaging.
MIL-STD-1388-2	-	DOD Requirements for a Logistic Support Analysis Record.
DOD-STD-1686	-	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment, Excluding Electrically Initiated Explosive Devices (Metric).

HANDBOOKS

MILITARY

MIL-HDBK-263	-	Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment, Excluding Electrically Initiated Explosive Devices (Metric).

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5904.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

AR 310-50	-	Catalog of Abbreviations and Brevity Codes.
AR 380-5	-	Department of the Army Information Security Program.
AR 380-380	-	Automation Security.
AR 385-11	-	Ionizing Radiation Protection (Licensing, Control, Transportation, Disposal, and Radiation Safety).
AR 385-62	-	Regulations for Firing Guided Missiles and Heavy Rockets for Training, Target Practice, and Combat.
TB 750-93-1	-	Functional Grouping Codes: Combat, Tactical, and Support Vehicles and Special Purpose Equipment.
TM 9-1300-206	-	Ammunition and Explosives Standards.

(Copies of other Government documents required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document ("This document" includes all volumes of MIL-STD-361 series.) and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Acronyms used in this standard. The acronyms used in this standard are defined as follows:

a. CPC - Corrosion Prevention Control.

b. DOD - Department of Defense.

c. DODISS - Department of Defense Index of Specifications and Standards.

d. EIR - Equipment Improvement Recommendation.

e. ESD - Electrostatic Discharge.

f. LSA - Logistic Support Analysis.

g. LSAR - Logistic Support Analysis Record.

h. MAC - Maintenance Allocation Chart.

i. NHA - Next Higher Assembly.

j. PCB - Printed Circuit Board.

k. PMCS - Preventive Maintenance Checks and Services.

l. QA - Quality Assurance.

m. RPSTL - Repair Parts and Special Tools List.

n. TAMMS - The Army Maintenance Management System.

o. TM - Technical Manual.

p. WTB - Warranty Technical Bulletin.

3.2 Caution. A statement or some other notification about an operating or maintenance procedure, practice, or condition that, if not strictly observed, could result in damage to, or destruction of, equipment or loss of mission effectiveness.

3.3 Chapter. The major, first-level functional division of a technical manual (TM).

3.4 Component. A constituent part not normally considered to be capable of independent operation; a piece part.

3.5 Contracting activity. The Department of Defense (DOD) component, activity, or organization of a using military service, or that organization delegated by a using service, that is responsible for the selection and determination of requirements for TMs.

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3.6 Equipment. One or more units capable of performing specified functions.

3.7 Multi-system equipment. Equipment with numerous sophisticated systems installed, including integrated systems and built-in test capability. Equipment falling into this category requires a high volume of technical information to adequately support operation and maintenance of the equipment. Such equipment includes, but is not limited to, installed systems, subsystems and aeronautical equipment, support equipment, and spare parts. The data shall be divided and arranged in TMs that contain comprehensive, usable information.

3.8 Note. A statement or some other notification that adds, emphasizes, or clarifies essential information of special importance or interest.

3.9 Section. The second-level functional subdivision of a TM and major division of a chapter.

3.10 System. A regularly interacting or interdependent group of items forming a unified whole to perform a specific operational function.

3.11 Technical manuals (TMs). Manuals that contain instructions for the installation, operation, maintenance, and support of weapon systems, weapon system components, and support equipment. TM information may be presented, according to prior agreement between the contractor and the Government, in any form or characteristic, including hard printed copy, audio and visual displays, electronic imbedded media, disks, other electronic devices, or other media. They normally include operational and maintenance instructions, parts lists, and related technical information or procedures exclusive of administrative procedures.

3.12 Unit. An assembly or any combination of spare and repair parts mounted together, normally capable of independent operation in a variety of situations.

3.13 Warning. A statement or some other notification about an operating or maintenance procedure, practice, or condition that, if not strictly observed, could result in long term health hazard, injury to, or death of personnel performing the task prescribed in the TM.

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4. GENERAL REQUIREMENTS

4.1 General style. This document sets forth introductory information and theory of operations requirements that shall be followed in the preparation and acquisition of TMs, revisions, supplements, and changes.

4.1.1 Conformance to standard. Unless requirement is specifically excluded by the contracting activity, the introduction chapter developed for a TM shall conform to this document and MIL-STD-361-1.

4.1.2 Technical content preparation. The following baseline requirements shall be used, as specified by the contracting activity:

a. Logistic support analysis record (LSAR). The technical data and instructions developed by the requirements of MIL-STD-1388-2, DOD Requirements for a Logistic Support Analysis Record (LSAR), shall be used as the baseline to prepare TMs.

b. Maintenance allocation chart (MAC). The MAC shall be used as the baseline to prepare TMs. The MAC shall be prepared from the equipment top-down breakdown and be in functional group code sequence to consolidate and identify those groups on the list which involve identified maintenance functions. Logistic Support Analysis (LSA) Output Report 04, Maintenance Allocation Summary will be used when available as source data for the final approved MAC.

When specified by the contracting activity and if a waiver of standard requirements is granted, TB 750-93-1 may be used as the baseline for preparing TMs.

4.1.3 Security classification markings. All classified TMs and material shall be safeguarded and identified in accordance with MIL-STD-361-1.

4.2 General information and warning summary.

4.2.1 General information. General information shall include a brief description of the equipment and reflect the following: type, series, purpose, limitations, and how, in general, the equipment accomplishes its mission. Appropriate illustrations and exploded views shall be included. If the TM describes more than one model, significant differences shall be explained.

4.2.2 Warning summary. Refer to MIL-STD-361-1.

4.3 Use of warnings, cautions, and notes. Refer to MIL-STD-361-1.

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4.4 Equipment and weapon systems configuration.

4.4.1 Configuration. System areas and compartments shall be described, and the system equipment and components contained in the areas shall be listed. The configuration description shall be supported by separate illustrations of each compartment and area, identifying the listed system equipment. (See figure 1.)

4.4.2 System equipment. The subsystems/equipment comprising the system shall be identified and described. Other equipment which is installed in the subject system compartments and areas need not be listed in the text or called out in the illustrations if they do not directly affect the operation or maintenance of the subject system. Descriptions of operator-attended equipment shall include general statements about the nature and purpose of the controls and indicators. The text shall be supported by illustrations. All equipment shall be shown, whenever possible, in relative-scale proportion. If necessary for proper support of the text, equipment shall be illustrated individually with significant features called out.

4.4.3 Associated-system equipment. As applicable, descriptions and illustrations of associated-system equipment shall be limited to the major units of that equipment. The descriptions shall be more condensed than those of subject system equipment; otherwise, the same requirements shall be applicable. In the descriptions, emphasis shall be placed on associated systems equipment that constitutes operational or functional interfaces with the subject system. Such units shall be included in the system illustrations.

4.5 System environment requirements. As applicable, the following information shall be provided:

a. A list of the environmental control requirements, such as limited temperature, humidity, or other limited conditions. Damage to be expected from exceeding these limits and procedures for minimizing the damage shall be given. These procedures shall give due consideration to the fact that the tactical situation may dictate system operation under conditions exceeding the design limits of the system.

b. A tabular summary of the effects of weather conditions on equipment that will affect system capability or cause equipment damage. This summary shall provide references to any special servicing procedures that must be accomplished because of climatic changes, such as adding antifreeze to coolants.

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5. DETAILED REQUIREMENTS

5.1 Front matter. The following subparagraphs contain necessary content information for front matter. Refer to MIL-STD-361-1 for front matter format requirements.

5.1.1 Front cover. The front cover of a standard TM shall be formatted and contain the information as outlined in MIL-STD-361-1.

5.1.2 Title block/reporting errors and recommending improvements. Refer to MIL-STD-361-1.

5.1.3 Table of contents. Refer to MIL-STD-361-1.

5.1.4 How to use this manual. This section shall be written to provide information and guidance to familiarize the user with any special or unusual features of the TM. Coverage shall lead the user through the TM and explain important features of the organization and content. For example, the format is explained; operating, troubleshooting, preventive maintenance checks and services (PMCS) are explained; and repair, maintenance instructions, and other pertinent information are explained. Any peculiarities in the basic arrangement of the TM shall be described. How-to-use information shall not repeat the instructions given within the chapters. If applicable, information shall include reference to and how to use the associated repair parts and special tools list (RPSTL) in conjunction with the TM.

5.2 Chapter 1 - introduction. This chapter shall provide the user with information for completing forms and records and familiarize the user with the equipment. Functional and physical descriptions of the major equipment, components, and applicable interface equipment shall be provided.

5.2.1 User orientation. At a minimum, Chapter 1 shall consist of the following sections:

Section I. General Information
Section II. Equipment Description and Data
Section III. Theory of Operation

5.2.2 Section I - general information. This section shall begin with a full-view illustration of the equipment showing features pertinent to the equipment user. A typical full-view illustration is shown in figure 2. This section shall include the following paragraphs, as applicable.

5.2.2.1 Scope. The scope paragraph shall contain a brief statement to tell what is covered in the TM. As applicable, the following information shall be included:

a. Type of manual (to be provided by the contracting activity).

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b. Model number(s) and equipment name(s) (to be provided by the contracting activity).

c. Purpose of equipment(s).

d. Special inclusions in the manual, such as drill procedures or on-vehicle loading plans.

5.2.2.2 Maintenance forms, records, and reports. This paragraph shall contain the following statement:

"Department of the Army forms and procedures used for equipment maintenance will be those prescribed by (as applicable) DA PAM 738-750, The Army Maintenance Management System (TAMMS); DA PAM 738-751, Functional Users Manual for the Army Maintenance Management System - Aviation (TAMMS-A); or AR 700-138, Army Logistics Readiness and Sustainability."

In addition, for conventional and chemical ammunition TMs, add the following statement:

"Accidents involving injury to personnel or damage to materiel will be reported on DA Form 285 (Accident Report) in accordance with AR 385-40. Explosives and ammunition malfunctions will be reported in accordance with AR 75-1."

5.2.2.3 Reporting equipment improvement recommendations (EIR). This paragraph shall contain the following statement:

"If your (insert equipment short item name) needs improvement, let us know. Send us an EIR. You, the user, are the only one who can tell us what you don't like about your equipment. Let us know why you don't like the design or performance. Put it on an SF 368 (Product Quality Deficiency Report). Mail it to the address specified in DA PAM 738-750. We will send you a reply."

5.2.2.4 Corrosion prevention and control (CPC). This paragraph shall contain a statement similar to the following:

Corrosion Prevention and Control (CPC) of Army materiel is a continuing concern. It is important that any corrosion problems with this item be reported so that the problem can be corrected and improvements can be made to prevent the problem in future items.

While corrosion is typically associated with rusting of metals, it can also include deterioration of other materials, such as rubber and plastic. Unusual cracking, softening, swelling, or breaking of these materials may be a corrosion problem.

If a corrosion problem is identified, it can be reported using SF 368, Product Quality Deficiency Report. Use of keywords such as "corrosion," "rust," "deterioration," or "cracking"

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will ensure that the information is identified as a CPC problem.

The form should be submitted to the address specified in DA PAM 738-750.

For aircraft TMs, this paragraph shall include a reference to TM 55-1500-343-23.

5.2.2.5 Destruction of Army materiel to prevent enemy use. This paragraph shall reference the appropriate TM(s) covering the destruction of Army materiel to prevent enemy use. The contracting activity will provide the TM number(s) to reference.

5.2.2.6 Preparation for storage or shipment. These requirements, including packaging and administrative storage, shall be part of the maintenance instructions, and reference to those instructions shall appear in this paragraph in Chapter 1.

5.2.2.7 Warranty information. When the TM covers equipment that is under warranty and a warranty technical bulletin (WTB) is published, the TM shall include only a reference to the applicable WTB. When a WTB is not published, pertinent warranty information shall be provided. An applicable statement such as the following shall be included:

The (insert name of equipment) is warranted for (insert miles or other timeframe as appropriate). The warranty starts on the date found in block 23, DA Form 2408-9 in the logbook. Report all defects to your supervisor, who will take appropriate action.

5.2.2.8 Nomenclature cross-reference list. An appropriate cross-reference list shall be included when non-official nomenclature (common name) is approved by the contracting activity (MIL-STD-361-1). See figure 3 for example. Once a common name, abbreviation, or acronym is used, it shall be used consistently throughout the TM.

5.2.2.9 List of abbreviations. A list of abbreviations, consisting of all abbreviations, acronyms, signs, or symbols used in the TM shall be included. For example, see figure 3. For aircraft TMs only, a statement shall be included that abbreviations are in accordance with MIL-STD-12 and AR 310-50, except when the abbreviation stands for a marking actually found in the aircraft.

5.2.2.10 Quality assurance (QA). Unless requirement is specifically excluded by the contracting activity, this paragraph shall be included and shall either reference the pertinent QA TM(s) or shall include the appropriate general QA information. If QA information is not referenced but is included in the TM, this paragraph shall state that the text of each quality assurance procedure paragraph or step in the TM is preceded (and highlighted) by the addition of "(QA)," and the abbreviation QA

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shall be explained either in the title of the paragraph or in the text. For aircraft maintenance TMs, refer to FM 1-511.

5.2.2.11 Safety, care, and handling. This paragraph shall cover general precautions and safety regulations.

a. For ammunition TMs, this paragraph shall include information to comply with AR 385-62. References to applicable ARs for range safety and danger zones during training and combat shall be included. Clear explanations and official definitions of such safety-related terms as misfire, hangfire, and cook-off, which describe characteristics associated with the specific items(s) covered by the TM under preparation, shall also be included. A reference to TM 9-1300-206 shall be made for general ammunition care, handling, and safety.

b. For TMs covering equipment with radioactive parts or components, a paragraph shall be included to comply with Nuclear Regulatory Commission provisions, and references to applicable ARs and safety documents on radioactive materials shall be included. If additional coverage on radioactive materials is needed, but is not included in applicable documents, such instructions shall be provided here as required. In addition, the following information shall be provided throughout the TM:

(1) Nuclear warning notices shall be placed at the beginning of any instruction covering procedures that will expose personnel to a nuclear radiation hazard.

(2) Procedures to be followed prior to maintenance actions, or in the event of breakage of radioactive parts or components, including safety, care, and handling instructions, shall also be included.

(3) Radioactive parts or components shall be shown and suitably identified on a parts location diagram or illustration, and suitable warning notices shall be included.

(4) When applicable, equipment data shall include a list of radioactive parts or components and the type and quantity of radioactive material involved.

(5) Instructions for the disposal of radioactive material, such as the requirement to double bag in plastic all broken tritium sources, shall be provided in accordance with AR 385-11.

c. Electrostatic discharge (ESD) control standards for the protection of electrical and electronic parts, assemblies, and equipment shall be described, when applicable. The ESD classes (see DOD-STD-1686 and MIL-HDBK-263) shall be identified. All ESD sensitive items shall be identified by class and the protective/control measures to be taken

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shall be given in instructions provided for such maintenance functions/tasks as removal, disassembly, inspection, repair/replace, assembly, testing, and packaging. Of primary importance in the identification of class 1 items requiring stringent protection are: (1) class 1 chips on a printed circuit board (PCB); (2) point of reduced or no protection due to absorption by a next higher assembly (NHA); and (3) change in an ESD requirement for a class 1 item resulting from installation of the item into its NHA.

5.2.2.12 Nuclear hardness. If equipment covered in the TM has nuclear survivability requirements (i.e., overpressure and burst, thermal radiation, electromagnetic pulse or transient radiation effects on electronics), (1) this paragraph shall so state; (2) applicable cautions shall be incorporated into the text to ensure that hardness of the equipment is not degraded during operation or maintenance; and (3) all hardness critical processes and steps shall be marked in accordance with MIL-STD-361-1. The following statement shall be included:

"All hardness critical paragraphs in this manual are marked with the symbol HCP as follows:

a. When an entire paragraph, including all subparagraphs, is considered hardness critical, only the major paragraph will be marked by the symbol HCP, placed between the paragraph number and the title.

b. When only certain processes and steps within a paragraph are hardness critical, only the applicable processes and steps will be marked by placement of the symbol HCP between each applicable step number and the text."

5.2.2.13 Security measures for electronic data. This paragraph shall include instructions for handling, loading, scrubbing, overwriting, or unloading classified electronic data under usual or unusual conditions. Instructions shall meet the requirements of AR 380-5 and AR 380-380 as they pertain to automation security.

5.2.2.14 Calibration. Equipment requiring calibration shall be indicated and reference shall be made to the publication containing the applicable calibration procedure.

5.2.3 Section II - equipment description and data. This section shall contain the information in the following paragraphs, as applicable.

5.2.3.1 Equipment characteristics, capabilities, and features. This information shall consist of an overall description of the equipment, including general capabilities, special features, and other like information (e.g., applications, limitations) which will be helpful in the operation and maintenance of the equipment. (See figure 4.)

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a. The equipment type, its portability or mobility, operational and special environmental features, and remote control features shall be stated.

b. Components and their functions shall not be described unless essential to continuity. For functional data, reference shall be made to theory of operation, when required.

c. When equipment covered varies in scope and application or has several applications within an end item, a brief explanation of the multiple usage shall be made and a simple diagram showing all aspects of a typical application shall be included.

d. For ammunition TMs, packing and packaging shall be described, including number of rounds per pack.

5.2.3.2 Location and description of major components. External and internal views of the equipment shall be used to show general features and all major components. The major components/items in the view shall be explained by the use of keyed text. (See figure 5.) The information contained in this paragraph shall not duplicate information in the paragraphs covering equipment characteristics, capabilities, and features and equipment data.

a. When the equipment is designed for use with other equipment, one or more diagrams shall be used to illustrate the use of the equipment. Only information pertaining to the user shall be included.

b. Location and contents of end item and major component identification plates shall be illustrated. Modification instruction, and warranty plates, stencils, or location of serial numbers shall be illustrated as necessary to the user.

5.2.3.3 Differences between models. Significant differences between equipment models or units of the same model that affect operator or maintenance actions shall be described, and the extent of interchangeability shall be indicated. These differences shall be related explicitly to equipment model, part number, or serial number ranges in such a manner that the TM user can identify the specific equipment configuration involved. When model differences exist but have no effect on operation or maintenance, this fact shall be stated. An example of a typical differences-between-models table is shown in figure 6.

5.2.3.4 Equipment data. Performance data, including numerical and other standard-related data applying to operational and maintenance functions shall be provided. The equipment data shall summarize the specific capabilities and limitations of the equipment and other critical data needed by the TM user for maintenance of the equipment. Vehicle and cargo space dimensions and metric and other equivalents shall be included where appropriate. See figure 7 for a typical example.

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5.2.3.5 Equipment configuration. For equipment which can be configured several different ways, information on the various possible configurations shall be provided.

5.2.4 Section III - theory of operation. This section shall contain a simple functional description of how the end item and major components work and interface. The description shall explain what functional effect operation and adjustment of controls, switches, and devices have on other components of the equipment.

a. Only absolutely essential information shall be provided which the TM user must know to operate and troubleshoot the equipment. If necessary, functional block diagrams shall be used to illustrate the interface and functional relationships between major components.

b. Description of components and assemblies which are maintained at higher levels shall be included only where necessary for clarity or continuity.

c. For multisystem equipment, theory of operation shall be organized so the first paragraph of the section contains an explanation of the end item as an integrated whole. Within each end item explanation, there shall be an explanation of the appropriate components; and within each component explanation, there shall be an explanation of the appropriate assemblies. Further breakdowns below the assembly level shall be included as appropriate.

For examples of theory of operation, see figures 8, 9, and 10.

5.3 Quality assurance requirements. Quality assurance and quality control for TMs, revisions, supplements, and changes shall be in accordance with MIL-STD-361-4.

5.4 Packaging and preparation for delivery. TMs, revisions, supplements, and changes shall be packaged in accordance with MIL-STD-361-5.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This document prescribes the introductory information with theory of operations requirements applicable to various types of technical manuals, revisions, supplements, and changes prepared by or for the Department of the Army.

6.2 Issue of DODISS. When this document is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation. (See 2.1.1.)

6.3 Tailoring guidance. The contracting activity should tailor any required options offered herein in accordance with MIL-M-63008.

6.4 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced. (See 2.1.1.)

c. Type of manual (5.2.2.1).

d. Model number(s) and equipment name(s) (5.2.2.1).

e. Destruction of Army materiel to prevent enemy use (5.2.2.5).

6.5 Supersession data. This document consolidates, standardizes, and supersedes the introductory information with theory of operation requirements contained in the following documents:

MIL-M-63036D(TM)	-	Manuals, Technical: Operator's, Preparation of.
MIL-M-63038C(TM)	-	Manuals, Technical: Organizational or Aviation Unit, Direct Support or Aviation Intermediate, and General Support Maintenance.
MIL-M-63041E(TM)	-	Manuals, Technical: Depot Maintenance Work Requirements.

6.6 Subject term (key word) listing. The following terms are to be used to identify this document during retrieval searches:

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Accident and safety reports
Acronyms used in this document
Corrosion prevention and control (CPC)
Differences between models
Equipment characteristics, capabilities, and features
Equipment configuration
Equipment data
General information
Handling and storage
Location and description of major components
Maintenance forms, records, and reports
Packaging and preparation for delivery
Preparation for storage or shipment
System environment requirements
Technical content requirements
Theory of operation
User orientation
Warnings and cautions
Warranty information

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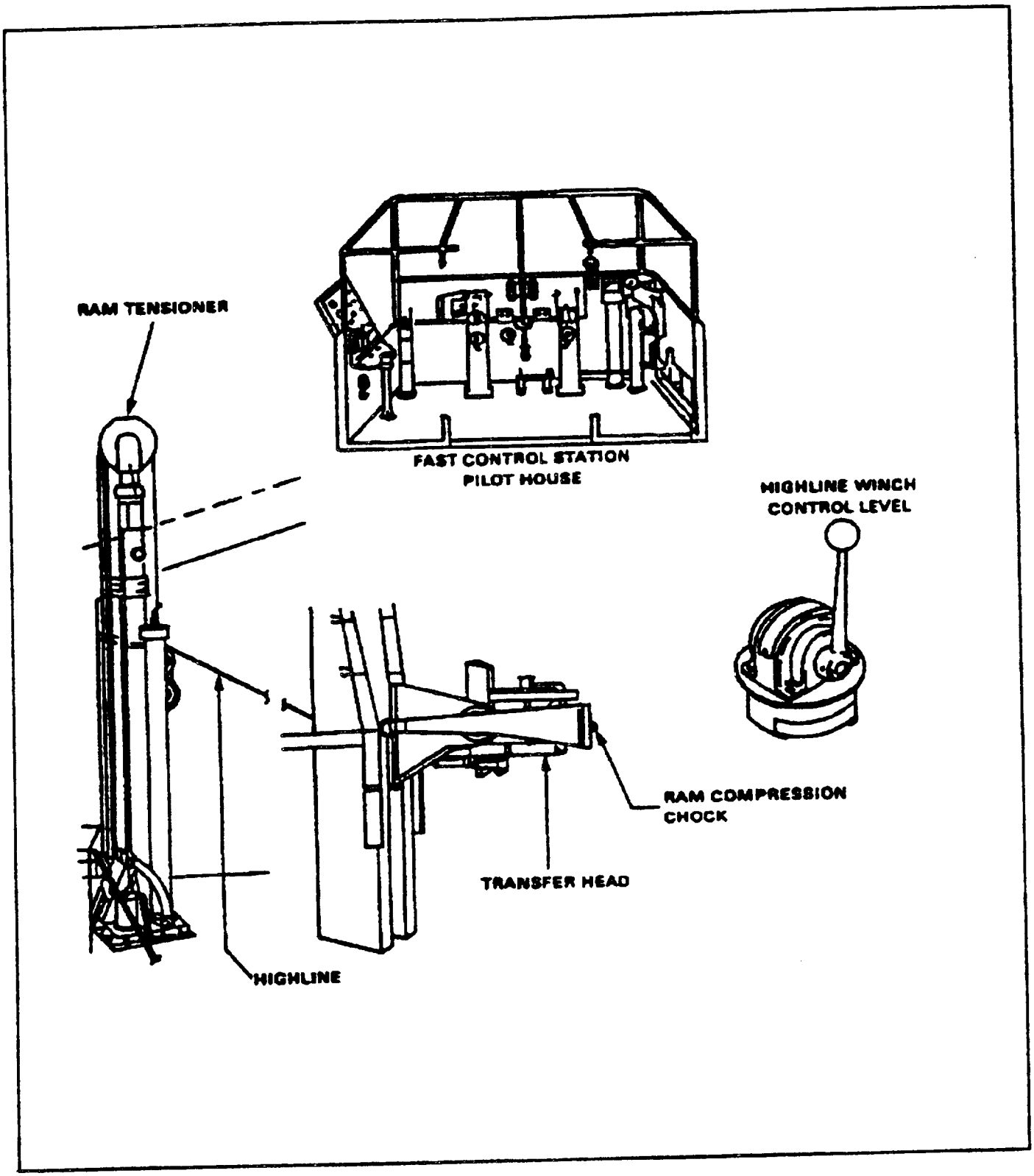


FIGURE 1. Example of compartment area illustration (reference 4.4.1).

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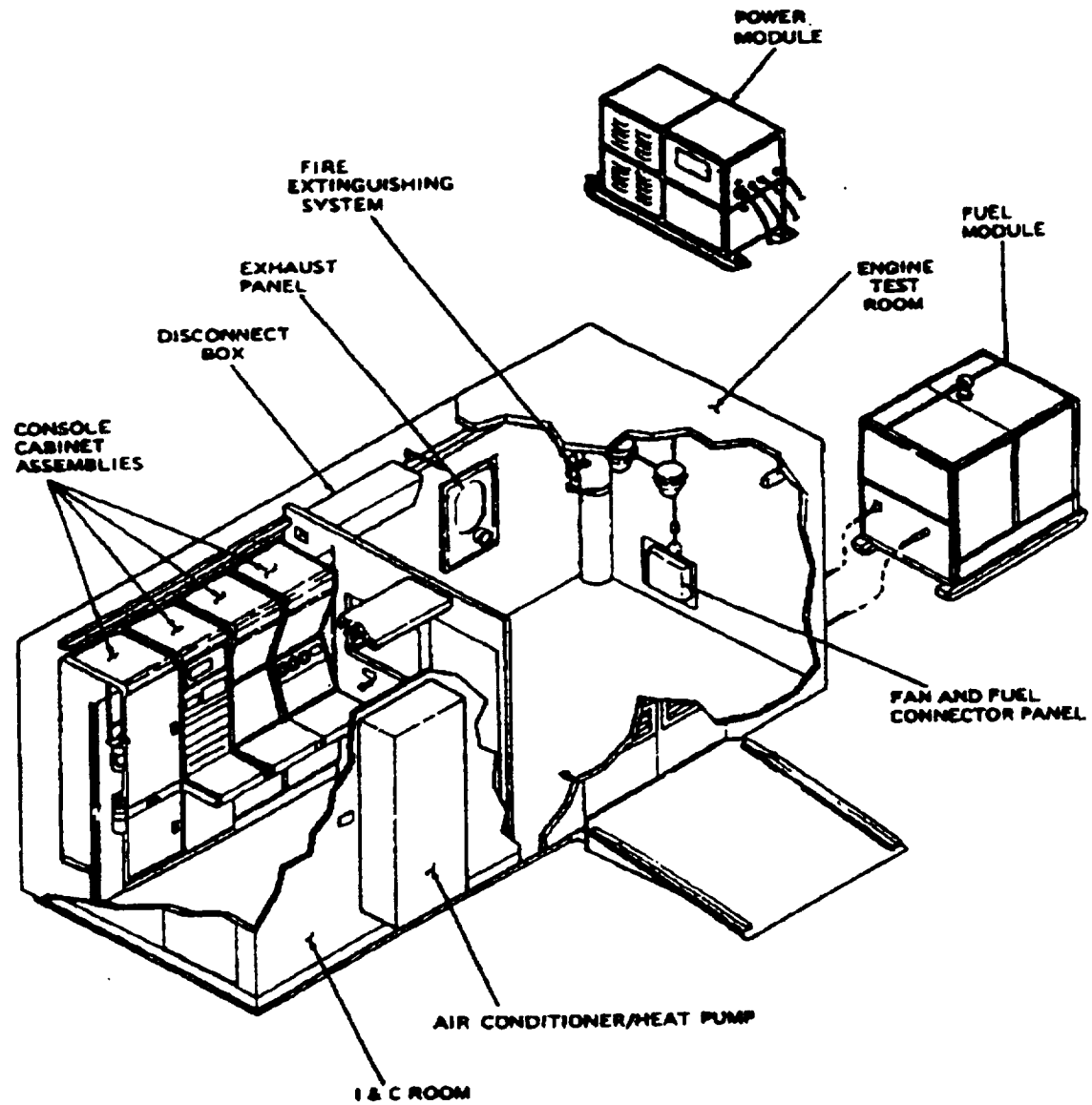


FIGURE 2. Example of full view illustration (reference 5.2.2).

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X.X NOMENCLATURE CROSS-REFERENCE LIST.

Common Name	Official Nomenclature
Fuel Gage	Liquid Level Indicator
Launcher (LZL)	Launcher, Zero Length, Guided Missile: M740
Loader Transporter (LT)	Launcher, Guided Missile, Carrier, Mounted: M752
Mobility Kit	Handling Unit, Guided Missile Equipment: M39

X.X LIST OF ABBREVIATIONS/ACRONYMS.

BDAR	Battlefield damage assessment and repair
CTA	Common table of allowances
HCP	Hardness critical process

FIGURE 3. Example of reference information (nomenclature cross-reference and abbreviations) (reference 5.2.2.8, 5.2.2.9).

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X.X EQUIPMENT CHARACTERISTICS, CAPABILITIES, AND FEATURES.

a. Characteristics.

Trailer has two wheels.

Each wheel has a service brake. Control of each brake is independent of the other.

Body is easily removed.

b. Capabilities and Features.

Maximum load: 1500 lb (681 kg)

Maximum speed with maximum load evenly distributed:

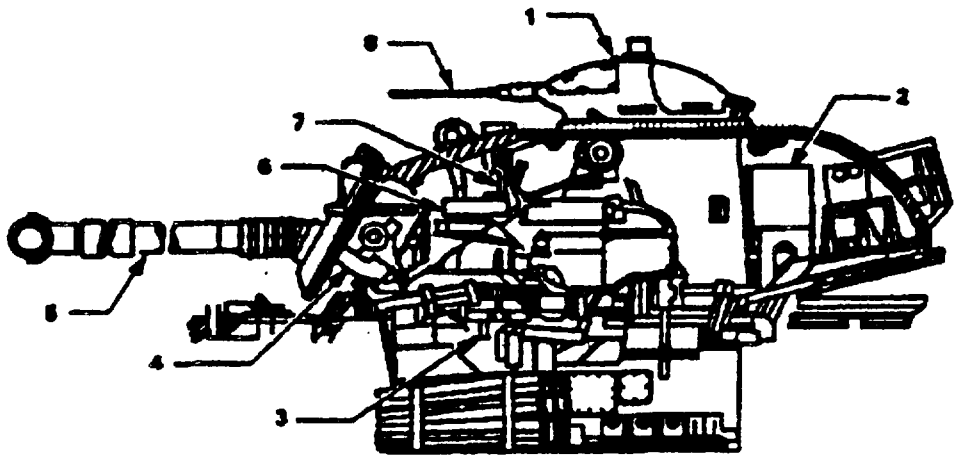
Highway: 50 mph (80 kph)
Cross-country: 30 mph (48 kph)

Cargo volume capability may be increased by installing tailgate and rack assembly.

Canvas cover assembly protects cargo from weather.

FIGURE 4. Example of a characteristics paragraph (reference 5.2.3.1).

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The turret is made from one-piece armor steel casting with openings for M87A1 combination gun mount, M1 commander's cupola, and external turret components. Rotates 360° to enable 90-mm gun, 7.62-mm machine gun, and caliber .50 machine gun to aim at targets in any direction.

1. M1 COMMANDER'S CUPOLA. Provides self-contained enclosure for commander with vision blocks for external vision. Contains caliber .50 machine gun and instruments and controls to operate machine gun.

2. COMMUNICATION COMPONENTS. Encloses ground-to-ground communication system units.

3. GAS PARTICULATE FILTER UNIT. Provides filtered air from 12-cfm filter through individual hoses to masks for commander, gunner, and loader.

4. M87A1 COMBINATION GUN MOUNT. Consists of recoil mechanism, mount, and attached gun shield. Pivots in elevation on turret trunnions. Supports 90-mm gun and 7.62-mm machine gun.

5. 90-MM GUN. Consists of quick-change tube, rectangular breeching, vertical-sliding breechblock, and breech mechanism. Includes bore evacuator and muzzle blast deflector. Fires electrically or mechanically.

6. M73 7.62-MM MACHINE GUN. Lightweight, air cooled, recoil operated, metallic link-belt fed with short receiver.

7. GUNNER'S CONTROLS AND INSTRUMENTS. Mechanical, electrical, and hydraulic controls and sighting components for 90-mm gun and 7.62-mm machine gun.

8. M2 CALIBER .50 MACHINE GUN, HB, TT. Automatic, recoil-operated, link-belt fed, air cooled, heavy barrel (HB), turret type (TT). Fires electrically or manually using solenoid trigger.

FIGURE 5. Example of location and description of major components illustration (reference 5.2.3.2).

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1.15 DIFFERENCES BETWEEN MODELS.

This paragraph shows major equipment and operational differences between models of M44A2 series vehicles. It has been organized in tabular form for easy reference by operators.

Table 1-1. Differences Between Models.

VEHICLE CHARACTERISTICS	M35A2	M35A2C	M36A2	M49A2C	M50A2	M50A3	M109A3	M185A3	M275A2	M342A2	M756A2	M764	DESCRIPTION (PARA NO.)
Body Features:													
Built-In A-Frame											X		2.31
Cab Protector										X	X	X	2.30,2.31,2.32
Earth Boring Machine												X	2.32
Floodlights											X		2.31
Nonreducible Height							X	X					1.28
Permanent Sides	X		X	X	X	X	X	X		X		X	--
Rear Winch											X	X	2.31,2.32
Reducible Height	X	X	X	X	X	X			X	X	X	X	1.28
Removable Sides		X									X		2.25,2.31
Operational Capabilities:													
Cargo/Personnel	X	X	X							X	X		2.25,2.30
Dump Operations										X			2.30
Earth Boring/ Polesetting												X	2.32
Equipment Repair							X	X					2.28
Fifth Wheel Operations										X			2.29
Fuel Servicing				X									2.26
Pipeline Construction											X		2.31
Water Servicing					X	X							2.27
Wheelbases:													
142 in. (360.6 cm)									X				--
154 in. (391.1 cm)	X	X		X	X	X	X	X		X	X	X	--
190 in. (482.6 cm)			X										--

FIGURE 6. Example of a table showing differences between models (reference 5.2.3.3).

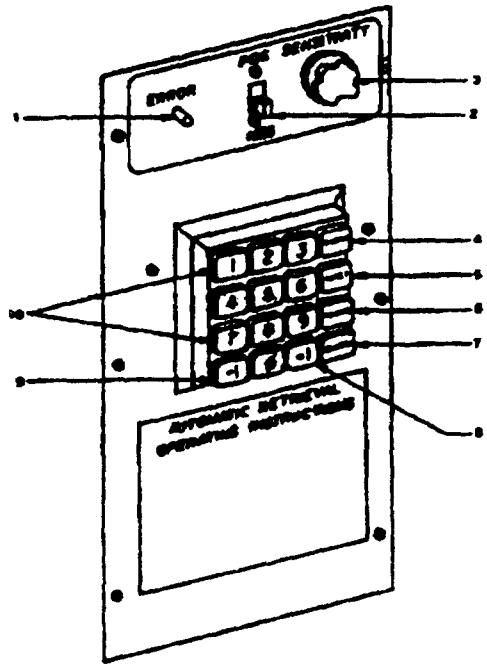
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Table X-X. Equipment Data.

WEIGHTS AND DIMENSIONS	
Ready for Travel	
Weight	1429 kg (3150 lb)
Length	472.4 cm (186 in.)
Width	200.7 cm (79 in.)
Height	205.7 cm (81 in.)
Maximum Towing Speed	72.42 kph (45 mph)
Emplaced with Cannon in Travel Lock	
Length	386.1 cm (152 in.)
Width	370.8 cm (146 in.)
Height	182.9 cm (72 in.)
Clearance (Traversing)	444.5 cm (175 in.)
Height (Cannon Full Elevation)	292.1 cm (115 in.)
Terrain Level	10° max slope
PERFORMANCE	
Firing Rate - Shots per Minute	
High Rate	3,000 spm
Low Rate	1,000 spm
Burst Limits - High Rate Only	10, 30 60, or 100 rd
RADAR RANGE	
Minimum	250 m (273 yd)
Maximum	5,000 m (5,468 yd) (1 sq meter target)

FIGURE 7. Example of equipment data table (reference 5.2.3.4).

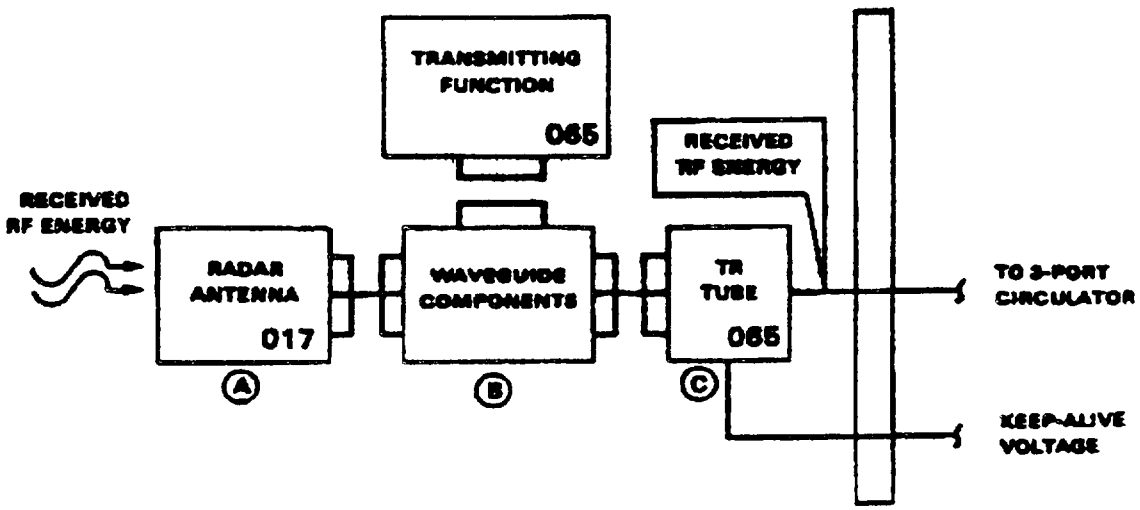
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INDEX NO.	CONTROL INDICATOR	FUNCTION
1	Error Indicator Light	When illuminated, indicates improper operation of film control assembly.
2	Positive/Negative Switch	Sets the equipment to read either positive or negative type microfilm.
3	Sensitivity Control	Provides adjustment of sensitivity of code sensing circuits used in automatic frame retrieval operation.
4	Clear Button	Clears all keyboard entries, extinguishes error light, and stops film movement when depressed.
5	Reset Button	Provides capability of setting microfilm starting frame number to number entered on keyboard.
6	Rewind Button	Provides motor drive microfilm rewind capability.
7	Search Button	Initiates frame retrieval when depressed.
8	+1 Button	Moves microfilm one frame forward when depressed.
9	-1 Button	Moves microfilm one frame in reverse when depressed.
10	0 thru 9 Buttons	Permits entering of desired frame number.

FIGURE 8. Example of theory of operation presentation for mechanical equipment (reference 5.2.4).

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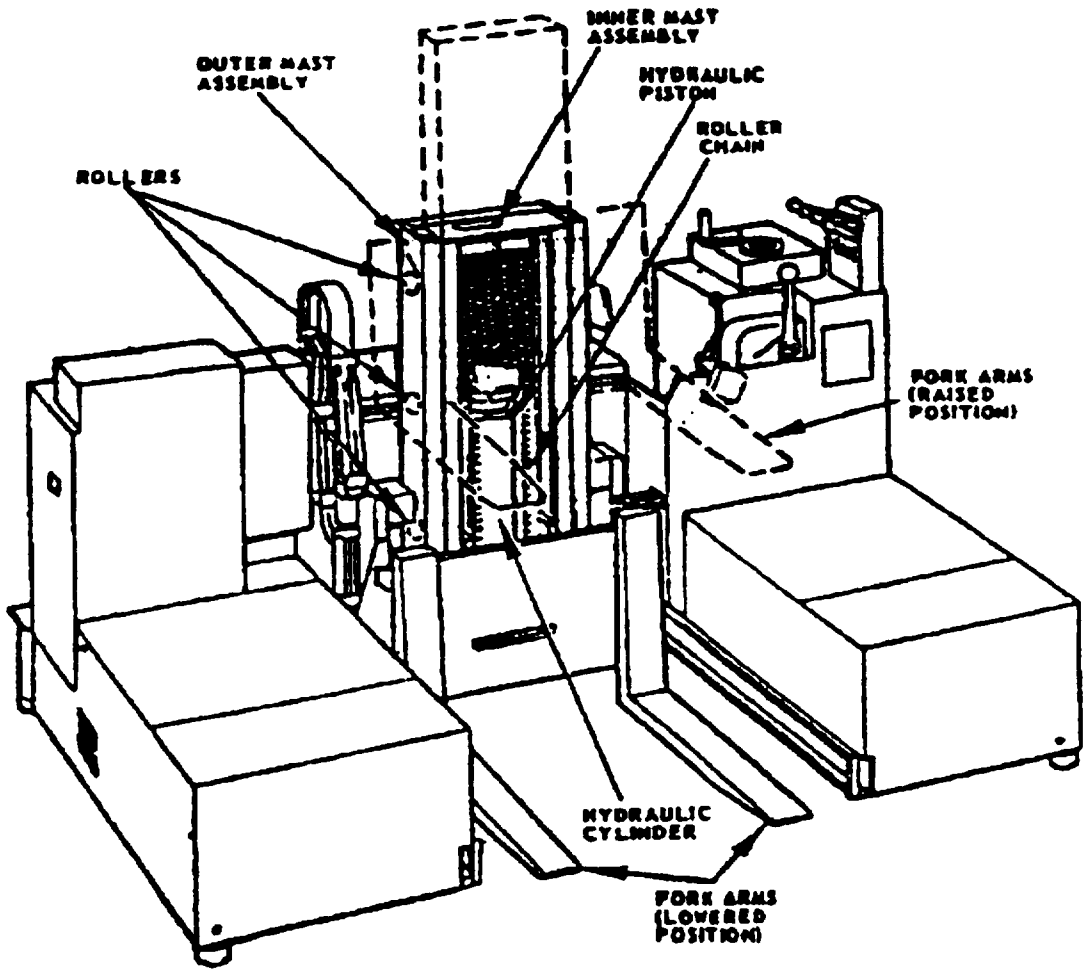
A. RADAR ANTENNA. Parabolic reflector which collects reflected energy from targets along antenna boresight axis. Mechanically positioned to search for and track targets. Modulates signal to develop azimuth and elevation pointing-error signals for positioning.

B. WAVEGUIDE COMPONENTS. Permit use of a single antenna for both transmitting and receiving. Circulate energy from transmitting function to antenna and from antenna to TR tube. Circulate any energy reflected from TR tube to resistive load.

C. TR TUBE. Short-circuits RF pulse leakage to prevent its entering receiving section when RF leakage may be of sufficient amplitude to damage paramp or image rejection mixer diodes. Keep-alive voltage maintains ionization during transmission but allows received weak signals to pass through tube without ionization. Voltage applied only during radar on operation.

FIGURE 9. Example of theory of operation presentation for electrical equipment (reference 5.2.4).

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a. Fork Assembly. The 48-inch long side reach forks are connected to the fork carriage. The fork carriage moves up and down by roller chains driven by a hydraulic cylinder and piston. The fork carriage is roller mounted to the inner mast assembly. Rollers mounted on inner mast move in channels which make up the outer mast assembly. As the fork carriage moves up, the top of the hydraulic piston assembly comes in contact with the inner mast. If the hydraulic piston moves up more, it lifts the inner mast along with the fork carriage. Forks can be raised from deck level to 60 inches above the deck by moving the fork carriage.

FIGURE 10. Example of theory of operation presentation for multisystem equipment (reference 5.2.4).

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3. DOCUMENT TITLE
Introductory Information with Theory of Operations Requirements

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

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a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
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c. ADDRESS *(Include Zip Code)* ATTN: AMXMD-EP (TMSS) Lexington, KY 40511-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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